Unit-05
Troubleshooting & Preventive Maintenance

❖ Requirement Of Computer Maintenance

- Troubleshooting in a computer is an intelligent′ and sophisticated process. Advances in IC integration have totally changed′ the nature of troubleshooting. however the original design of computer is comes′ from IBM PC and intel's microprocessor. Success of troubleshooting is depends on:

1. Tolls and test equipment used′
2. Diagnostic aids in the PC′
3. Problem solving approach′
4. Various techniques adopted in troubleshooting′

❖ What is POST(Power-On-Self-Test)?

- A power-on self-test (POST) is a set of routines performed by firmware or software immediately after a computer is powered on, to determine if the hardware is working as expected.
- The process would proceed further only if the required hardware is working correctly, else the BIOS(Basic Input Output Software) would issue an error message.
- POST sequence is executed irrespective of the Operating System and is handled by the system BIOS.
- Once the tests are passed the POST would generally notify the OS with beeps while the number of beeps can vary from system to system. When POST is successfully finalized, bootstrapping is enabled.
- Bootstrapping starts the initialization of the OS.

❖ The Role of POST in the Boot Sequence

- The boot sequence is the process of starting a computer/system.
- The boot process is initiated when the power button is pressed, it sends power to the boot-loader in the cache memory.
- The Boot loader performs POST as a preboot sequence and if everything is working well without any errors the BIOS(Basic Input Output System) is activated which finds and loads the operating system.
- Finally the software has to interact with the hardware units to complete the process. To avoid any hardware errors while executing a software program, the pre-boot sequence would test the hardware and initiate the OS if and only if the basic hardware units are functioning as expected.
- He principal duties of the main BIOS during POST are as follows:
    1. Find, size, and verify the system main memory.
    2. Initialize BIOS.
    3. Identify, organize, and select which devices are available for booting.
    4. Verify CPU registers.
    5. Verify the integrity of the BIOS code itself.

6. Verify some basic components like DMA, timer, interrupt controller.
7. Pass control to other specialized extensions BIOS (if installed).

➢ **The checks are performed majorly on:**

- Hardware elements like processor, storage devices and memory.
- Basic System Devices like keyboard, and other peripheral devices.
- CPU Registers
- DMA (Direct Memory Access)
- Timer
- Interrupt controller

❖ Errors about the Power-on Self-test

- Almost everything that might prevent the computer from continuing to boot will signal some kind of error.
- Errors can come in the form of flashing LEDs, audible beeps, or error messages on the display, all of which are technically called POST codes, beep codes, and on-screen power-on self-test error messages. The following chart is about beeps and their corresponding meanings:

Beeps	Meaning
1. Short beeps	Normal POST – the system is OK
2. Short beeps	POST error – error code is shown on screen
3. No beep	Power supply, system board problem, disconnected CPU, or disconnected speaker
4. Continuous beep	Power supply, system board, or maybe RAM problem, keyboard problem
5. Repeating short beeps	Power supply or system board problem or keyboard
6. 1 long, 1 short beep	System board problem
7. 1 long, 2 short beeps	Display adapter problem (MDA, CGA)
8. 1 long, 3 short beeps	Enhanced Graphics Adapter (EGA)
9. 3 long beeps	3270 keyboard card

❖ Explain in detail Power on Self Test (POST).

- POST is the common term for the computer, router or printer's pre-boot sequence.
- The same basic sequence is present on all computer architectecture. it is the first of the more general process called **initial program load (IPL)** , booting or bootstrapping.
- When the problem is identified with the system during the POST, the BIOS will use three methods to represent the problem.

A. Beep Code: Beeping pattern on the speaker.
Example: five beep means, processor may be fail.

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B. **Error code:** for every error there is specific code which is describing the nature of error. This code description is provided in the specification of the BIOS manufacturer.
Example: Error code from 100 to 199 indicates CPU related fault.

C. **Detail Error Message:** Brief description of the error is display on the screen.
Example: INTR#1 means interrupt controller 1 failed.

The following procedure describes POST sequence:

1. **Processor test**
   Different flags and registers within the processor are tested.
2. **BIOS ROM test**
   The contents of the 8 k ROM containing POST and BIOS is verified by checksum calculation method.
3. **Timer 1 test**
   The timer 1 in programmable interval time (8253-5) is tested.
4. **DMA channel 0 test**
   The channel 0 of DMA controller is tested here.
5. **Base 16k RAM test**
   The first 16k RAM is tested using five different test patterns are written and verified on each location of the address 00000h to 03FFFFh.
6. **CRT controller test**
   CRT controller (6845) and the video buffer RAM in the display adapter are tested.
7. **Motherboard support chips test**
   The interrupt controller, timer (8253) and keyboard interface are tested.
8. **RAM test**
   RAM after the first 16kb tested here using five different patterns.
9. **Optional ROM test**
   The ROMs in Hard disk controller are tested by checksum method.
10. **Peripheral controller test**
    The floppy controller, parallel ports, serial ports and Hard disk controller are tested.
    At end of all the above tests, the POST passes control to the bootstrap program in BIOS. The bootstrap loader reads the initial program from track 0 on disk. This initial program reads more programs from the disk, which is nothing but the operating system.

❖ **Error messages with code**

Code	Description
100 to 199	System boards
200 to 299	Memory
300 to 399	Keyboard
400 to 499	Monochrome display
500 to 599	Color/ graphics display
600 to 699	Floppy- disk dive or adapter
700 to 799	Math coprocessor
900 to 999	Parallel printer port
1000 to 1099	Alternate printer adapter

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1100 to1299	Asynchronous communication device, adapter , or port
1300 to 1399	Game port
1400 to 1499	Color / graphics printer
1500 to 1599	Synchronous communication device, adapter, or port
1700 to 1799	Hard drive and / or adapter
1800 to 1899	Expansion unit (XT)
2000 to 2199	Bisynchronous communication adapter
2400 to 2599	EGA system-board video (MCA)
3000 to 3199	LAN adapter
4800 to 4999	Internal modem
7000 to 7099	Phoenix BIOS chips
7300 to 7399	3.5-inch disk drive
8900 to 8999	MIDI adapter
11200 to 11299	SCSI adapter
21000 to 21099	SCSI fixed disk and controller
21500 to 21599	SCSI CD-ROM system

❖ BIOS Beep Code and Error Message:

Beep code	Error Message	Description
1 long , 2 short	Video adapter error	Either video adapter is bad or is not seated properly. Also, check to ensure the monitor cable is connected properly.
Repeating (endless loop)	Memory error	Check for improperly seated or missing memory.
1long, 3 short	No video card or bad video RAM	Reseat or replace the video card.
High frequency beeps while running	Overheated CPU	Check the CPU fan for proper operation. check the case for proper air flow.
Repeating High / Low	CPU	Either the CPU is not seated properly or the CPU is damaged. May also be due to excess heat. check the CPU fan or BIOS setting for proper fan speed.

❖ Meaning of troubleshooting

- Troubleshooting is a systematic approach to problem-solving that is often used to find and correct issues with complex machines, electronics, computers and software systems.

❖ Basic computer troubleshooting steps

1) My computer freezes or is behaving strangely

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- Try restarting your computer. Many basic problems can be resolved easily and quickly this way.
- Press the Ctrl & Alt & Del keys on your keyboard together at the same time. This should bring up a menu that will allow you to run Task Manager. In Task Manager, switch to the Applications tab. Highlight any programs with the status 'Not Responding' and choose End Task.
- You may be asked to confirm if you want to end the unresponsive program, so choose Yes. Do this for all programs that are not responding.
- If all else fails and you cannot shutdown/restart your computer, then hold down the power button on the machine until it forcibly turns off. Wait a few seconds and then turn it back on again.

2) My computer doesn't power up

- Check that all the cables are securely plugged into the back of the machine and the monitor.
- Check that the power cables are plugged into a power socket and the socket has been turned on.
- Try using a different power socket or, if you are using a power extension strip, plug the power cable directly into a power socket in the wall.
- Replace the power cable with one that you know works.
- Check if there are any lights on at the front of the machine: If there are lights on the machine but not the monitor, then it's probably a monitor issue.If there are lights on the monitor but not the machine, then it's probably a machine issue. If there are no lights on anything, then it may be possible there is a local power cut.
- With laptops, try removing the power cable and the battery. Hold down the power button for about ten seconds, and then plug the battery and power cable in again. Press the power button to see if it switches on.

3) Nothing appears on the monitor

- Make sure both the computer and monitor are on.
- Make sure the monitor is securely plugged into the computer.
- Make sure the power cable is firmly plugged into the monitor.
- Some computers have multiple display ports, so make sure you have plugged the monitor into the correct one. Try each one in turn, switching the monitor off then on in between moves.
- Most monitors have a status window displayed when you turn it on. Check if you can see this status window when you press the power button on the monitor. You can also try this with the menu button on the monitor, which should bring up an options menu on the screen. This shows the screen is working ok, so it may be an issue with the video cable from the monitor or the machine itself.
- Check the brightness & contrast levels of the monitor via the menu button, to make sure it has not been set too dark.
- Move the mouse or press any key on the keyboard to make sure the screensaver hasn't activated or that the computer hasn't gone into standby/ hibernation mode.

4) Keyboard / Mouse doesn't work

- Make sure the keyboard/mouse is firmly plugged into the back of the computer
- Try unplugging one or both, and then reinserting it into the back of the computer
- Try plugging your USB keyboard/mouse into a different USB socket
- Replace the keyboard/mouse with one that you know works

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- If you cannot see any lights on your keyboard when you press the Caps Lock or Num Lock key, it may be a dead keyboard
- Make sure there is no dirt or fluff clogging up either the optical laser or roller ball on the underside of your mouse or keyboard. It may require a clean
- If you are using a wireless keyboard/mouse, try pressing the reset button on the device or replace the batteries.

5) Run fewer program at the same time

- Don't have too many programs running at the same time. Each running program consumes a bit of the system's resources. Have multiple open windows for the same program (e.g. having three Word documents open) also lowers resources as each window takes up a bit of memory and processing power.
- If you are not using an open file or program, close it so that the files/programs you are working in can speed up a little.

6) Restart your computer

- If you haven't restarted your computer for a while, or your normally put it into Sleep or Hibernation modes, allow it to restart. This clears the memory and allows your system to 'clean boot', freeing up resources in the process.

7) Remove viruses and malware

- Viruses and malware can slow down your computer, as well as causing other problems like not allowing you to access the internet or programs installed on your computer.
- Make sure you have antivirus software installed on your computer, and that it is enabled and up to date. Then run a full system scan to remove anything suspicious from your computer.
- Windows 10 includes Windows Security, which provides the latest antivirus protection. Your device will be actively protected from the moment you start Windows 10.
- Windows Security continually scans for malware (malicious software), viruses, and security threats. In addition to this real-time protection, updates are downloaded automatically to help keep your device safe and protect it from threats.
- You can customize how your device is protected with these Windows Security features. To access them, select Start > Settings > Update & Security > Windows Security .
- Then select the feature you want to explore. You can also use a free dedicated malware scanner such as Malware byte Anti-Malware to scan your computer. Download, install and run it.
- Allow it to update itself, and then choose 'Perform full scan' At the end of the scan click on the 'Remove selected' button to remove the malware. You may need to restart your computer.

❖ **Troubleshooting by Visual Inspection**

- Improperly set controls and loose or improperly connected cables are common causes of problems with hardware components. When investigating a system problem, first check all the external switches, controls, and cable connections. See Performing an External Visual Inspection.
- If this does not resolve your problem, then visually inspect the system's interior hardware for problems such as a loose card, cable connector, or mounting screw. See Performing an Internal Visual Inspection.
- **Performing an External Visual Inspection**

  1. Turn off the system and any attached peripherals (if applicable).

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2. Verify that all power cables are properly connected to the system, the monitor, and peripherals, and check their power sources.
3. Inspect connections to any attached devices including network cables, keyboard, monitor, and mouse, as well as any devices attached to the serial port.

- **Performing an Internal Visual Inspection**

1. Shut down the operating system, if necessary, and turn off the platform power on the front of the workstation.
2. Turn off the AC power on the rear of the workstation.
3. Turn off any attached peripherals, but do not disconnect the power cables.
4. Remove the left-side panel, following the procedures in
5. Verify that the components are fully seated in their sockets or connectors and that the sockets are clean.
6. Verify that all cables inside the system are firmly attached to their appropriate connectors.
7. Replace the left-side panel.
8. Reconnect the system and any attached peripherals to their power sources, and then power them on.

❖ **What is Systematic Troubleshooting procedure?**

- Troubleshooting is a step-by-step process. Below, we break it down into six simple to follow steps. It doesn't matter if you are an advanced or inexperienced professional; you will follow the same systematic approach every time.

Step 1: Define the problem

- The first step of solving any problem is to know what type of problem it is and define it well. A clear definition is fundamental when troubleshooting.
- When looking at a problem, you need to know what you are up against and the possible causes. Is it machine failure, an unexpected operation, user error, or a random anomaly? What happened that alerted you to the problem?
- Some equipment will have built-in ways of letting you know; alarms can sound, red lights flash, or a warning can go off when certain parts overheat. These signals can help with problem-solving. Other equipment just stops working.
- Whatever the case may be, you have to identify and define the problem before you can move forward.

Step 2: Collect relevant information

- You need to gather all the available information about the machine and its operations. You'll need the machine manual, any data regarding operations. For example, how often the machine is used, by whom, for what, and how long. You will also need the maintenance history, problem reports, etc.
- A modern CMMS like Limble should have the option to keep a digital copy of all the documents, history, and information. If communication with the Original Equipment Manufacturer (OEM) is possible, the maintenance crew can discuss the issue first. Sometimes calling the OEM is the fastest and easiest way to get the right help.

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Step 3: Analyze collected data

- Using all the information you have gathered, available checklists, and as much technical know-how as you can muster, you can now try to determine the root cause of the problem. Seek out expertise from other maintenance trouble-shooters or the person who reported the fault. It's much easier to solve a problem that you have seen before.

Think about recent changes to the asset. Ask yourself:

- Did we use new replacement parts?
- Has there been an upgrade lately?
- Did we change the type of input material we use?
- Has the device been used in a different way than usual?
- Has there been an electrical surge?
- Recent changes to the system or environment can often explain why the problem has come up.

- If you still have no clue what caused the problem after analyzing the data, you need to go back to Step 2 and collect more info. It is possible to overlook things or disregard something as unimportant during the first round of the information-gathering process.
- After this exercise, the person performing troubleshooting should form an educated guess and put forward some solutions.

Step 4: Propose a solution and test it

- Using what you know from above, you can create your plan of attack. You will get to the solution through a process of elimination and trial and error. In some cases, you may be able to test your theory on a smaller scale asset. You may have multiple options to try. Start with the simplest one first and work from there.

Take the following into account:

- potential safety concerns
- all the required resources and associated costs
- how complex the implementation will be
- the long-term outlook for the machine
- any personal biases person performing the troubleshooting may have
- Keep testing until you are sure that you have found the right solution. If nothing works, you will need to rethink what the actual cause is.

Step 5: Implement the solution

- Once you have accurately diagnosed the problem, found the solution, and tested it, it's time to get your hands dirty and fix it. Even if your solution worked during testing, it is important to test it again. Ensure

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the asset is working the way it should before you pack up and sign off. You'll also want to make a note of all the steps you take as you make them, so you don't forget what you've done.

❖ Various Peripheral Devices Troubleshooting:

1) Motherboard:-

- The motherboard and processor are two of the most important hardware components inside the computer.
- The various pieces of hardware inside the PC communicate with one another through the circuits on the motherboard, while the CPU stores and executed programming instruction.
- The motherboard and CPU can both be expensive to replace, however, but diagnosing hardware failure on your own can reduce the potential repair costs. Hardware components exhibit symptoms when falling.

1. **System Board Failure**
   Cause: DMA or timers.
   Solution:
   - Clear CMOS.
   - Remove expansion boards.
   - Replace the system board.
2. **Time & Date Not Set**
   Cause: Invalid time or date in configuration memory. RTC (real-time clock) battery may need to be replaced.
   Solution: Reset the date and time under control panel. If the problem persists, replace the RTC battery.
   Cause: CMOS jumper may not be properly installed.
   Solution: Check for proper placement of the CMOS jumper if applicable.
3. **Memory Size Error**
   Cause: Memory configuration incorrect.
   Solution:
   - Run Computer Setup or Windows utilities.
   - Make sure the memory modules are installed properly.
   - Verify proper memory module type.
4. **Memory Error**
   Cause: RAM failure.
   Solution:
   - Run Computer Setup or Windows utilities.
   - Ensure the memory modules are installed properly.
   - Verify proper memory module type.
   - Remove and replace the identified faulty memory module.
   - If the error persists after replacing memory modules, replace the system board.

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5. **Display Adapter Failure**
   Cause: Graphics display controller.
   Solution:
   - Reset the graphics card
   - Clear CMOS.
   - Verify monitor is attached and turned on.
   - Replace the graphics card
6. **CPU Fan not detected**
   Cause: CPU fan is not connected or may have malfunctioned.
   Solution:
   - Reseat CPU fan.
   - Reseat fan cable.
   - Replace CPU fan.
7. **System has no power at all. Power light does not illuminate, fan inside the power supply does not turn on , and indicator light on keyboard does not turn on.**
   Cause: Power cable is unplugged.
   Solution : Make sure power cable is secured plugged in.
   Cause: Defective power cable
   Solution : Replace cable
   Cause: Power supply failure.
   Solution: Power cable and wall socket are OK, but system is still dead. Replace cable.
   Cause: Faulty wall outlet; circuit breaker or fuse blown
   Solution: Use different socket, repair outlet, reset circuit breaker or replace fuse.
8. **System only boots from Floppy Disk. Hard Disk can be read and applications can be used, but booting from Hard Disk is impossible.**
   Cause: Hard Disk boot program has been destroyed.
   Solution: Back up data and applications files.
9. **Screen message says " Invalid Configuration" or " CMOS Failure."**
   Cause: Incorrect information entered into the configuration program.
   Solution: Check the configuration program. Replace any incorrect information. make sure correct information is in setup.
10. **Screen is blank**
    Cause: No power to monitor. Power connectors may be loose or not plugged in.
    Solution: Check the power connectors to monitor and to system. Make sure monitor is connected to display card, change I/O address on network card if applicable.
11. **System does not boot from hard disk can be booted from floppy disk drive.**
    Cause: Connector between hard drive and system board unplugged.
    Solution: Check cable running from disk to disk controller on the board. Make sure both ends are securely plugged in; check the drive type in the Standard CMOS setup.
12. **Floppy drive lights stay on.**
    Cause: Floppy Drive cable not connected correctly.

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Solution: Reconnected floppy cable making sure PIN1 on the Floppy Drive corresponds with PIN1 on floppy cable connector.

13. **C : drive failure.**
    Cause: Setup program does not have correct information.
    Solution: Boot from drive A: using DOS system disk. Input correct information to SETUP program.
    Cause: Hard Drive cable not connected properly.
    Solution: Check Had drive cable.
14. **Cannot boot system after installing second hard drive.**
    Cause: Master / Slave jumpers not set correctly.
    Solution: Set master / Slave jumpers correctly.
15. **Missing operating system on hard drive.**
    Cause: CMOS setup has been changed.
    Solution: Run setup and select correct drive type.

2) Keyboard:

1. **Keyboard won't respond anything**
   Causes and Solutions:
   - Check the keyboard connection to the computer and restart the computer.
   - Check the keyboard driver is properly installed or not. That we can seen in the device manage utility program in windows environment.
   - Verify the connection between controller and motherboard.
2. **Receiving a keyboard error when system is boot**
   Causes :
   - A key being pushed down while the computer starts.
   - Something on top of the keyboard.

   Solutions:
   - Make sure you do not boot the PC with a key pressed down on the keyboard.
   - Make sure nothing is on top keyboard during boot time.
3. **Pressing {Ctrl} + {Alt}+{Delete} keys, system does not respond**
   Causes :
   - Any one of the key mechanism is damaged.
   - System having virus.

   Solutions :
   - Individually test each of key.
   - Open other program and pressed the keys.
   - Scan our system with best antivirus software.
4. **One or more key on keyboard not generating a keystroke reliably or key are or misbehaving.**
   Causes :

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- Particular key is used frequently (gamming program)
- Keyboard having dirt and debris.
- If key internal mechanism is damaged.

Solutions :

- Clean the keyboard carefully
- Verify key on the other program.
- Change the key or keyboard.

5. **Keys are repeating**

Cause :

- BIOS typematic delay setting is not proper

Solutions :

- Check in BIOS for a setting to control the typematic delay and increase the repeating time

6. **Keyboard is locked, no keys function.**

Cause and solution:

- Keyboard is locked. Unlock keyboard.

7. **Keyboard not found message.**

Cause and solution:

- Your keyboard is not plugged into the computer securely. Unplug it and plug it back in and the problem should go away.

8. **Keys Don't Work.**

Cause and solution:

- If one of the keys on your keyboard gets stuck, turn the computer off and clean with a damp cloth.
- If only some of the keys on your keyboard are not working, the first would be to check the device status and update the keyboard's drive.

9. **Keyboard number don't type.**

Cause and Solution:

- There is a number lock button near your keypad that turns the pad on and off. If your numbers won't type, you have probably pressed this button by mistake so on the number lock key.

10. **The keyboard output is very slow or some characters are lost when typing .**

Causes and Solution:

- Wrong configuration
- Change the output speed of the keyboard.

3) Hard Disk Drive:

1. **Hard disk will not auto detect in the CMOS setup program**

Causes and Solutions :

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- Try hard disk in other system; if problem will not hard disk have problem, otherwise problem in system. Or try with Hard disk if problem not solve then system have some problem otherwise previous Hard disk have problem
- Verify Hard disk drive get enough power and is spinning. If it does not spin, then drive has problem or power problem.
- Make sure the power input connection to the disk is tight.

2. **Computer is running slowly or opening programs on the hard disk has become slow.**

Cause :

- The volumes on your hard disk may have become excessively fragmented, possibly because you have recently created a large number of files or folders or have installed new software.

Solution :

- Analyze and then defragment the volumes on your hard disk if necessary. For more information

3. **Hard disk drive is not detected.**

Causes :

- Miss alignment of power/interface cable
- Wrong setting of Master/Salve Jumper

Solutions :

- Verify the Hard disk drive power/interface cable
- Verify the Master/ Salve Jumper setting.

4. **The Hard disk drive is not bootable nor accessible at all**

Causes :

- Wrong Configuration
- Hard disk is dead

Solutions :

- Detect you Hard disk in CMOS setup, if it detect then there is operating system is crash, Install operating system.
- Verify the Master/Salve Jumper setting.

5. **The Hard disk drive generate more noise**

Solutions :

- Verify the hard disk drive is the source of the noise by removing the cover of the system and identifying the hard disk drive as the source of the noise.
- Run diagnostics on the hard disk drive. Replace drive if diagnostic testing reveals a failures.
- Back up your important data, it may be dead soon!

6. **"No ROM basic system halted" error message during start up (there are no active partitions)**

Causes :

- Drive is not install properly
- There is no active partition

Solutions :

- Verify that drive is install properly Set active partition using FDISK or other utility program.

7. Hard drive error occurs:

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Cause: Hard disk has bad sectors or has failed.
Solution: Select a drive. Under Error-checking, click check now. Use a utility to locate and block usage of bad sectors. If necessary , reformat the hard disk.

8. **Disk transaction problem.**
   Cause: Either the directory structure is bad or there is a problem with a file.
   Solution: Select a drive. Select file > Properties > tools. Under error –checking, click check now.
9. **The system error message, " HDD controller failure" appears:**
   Cause and solution:
   - Confirm the jumper setting on the drive.
   - Verify the drive type and make sure the drive spins up.
10. **Drive failure:**
    Cause and solution:
    - SETUP program does not have correct information. Boot from drive A : using DOS system disk. Input
    - Correct information to SETUP program.
    - Hard drive cable not connected properly. Check hard drive cable.

4) Printer :

1. **An application document does not print.**
   Cause: There is a document error in the print queue.
   Solution: Manage the print jobs by cancelling the document from the print queue and print again.
2. **Printer cannot be added or there is a print spooler error.**
   Cause: The printer service is stopped or not working properly.
   Solution: Start the print spooler and if necessary reboot the computer.
3. **Printer jobs are sent to the print queue but are not printed.**
   Cause: The printer has been installed on the wrong port.
   Solution: Use printer properties and settings to configure the printer port.
4. **Print queue is functioning properly, but the printer does not print.**
   Cause: There is a bad cable connection.
   Solution: Check for bent pins on the printer cable and check the printer cable connections to the printer and computer.
5. **Printer prints unknown characters or does not print anything.**
   Cause: Incorrect print driver installed.
   Solution: Uninstall incorrect print driver and install correct driver.
   Cause: Printer cables are loose.
   Solution: Secure printer cables
   Cause: No paper in printer.
   Solution: Add paper to the printer.
6. **Paper jams when printing.**
   Cause: Printer is dirty.
   Solution: Clean the printer

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Cause: The wrong paper type is being used.
Solution: Replace paper with the manufacturer's recommended paper type.
Cause: Humidity causes the paper to stick together.
Solution: Insert new paper in the paper tray.

7. **The toner is not fusing to the paper.**
   Cause: The toner is not fusing to the paper.
   Solution: Replace the toner cartridge.
   Cause: Toner cartridge is defective.
   Solution: Replace the toner cartridge.
   Cause: The paper is incompatible with the printer.
   Solution: Replace the paper.
8. **The paper is creased after printing.**
   Cause: The paper is defective.
   Solution: Remove the paper from printer, check for defects, and replace.
   Cause: The paper is loaded incorrectly.
   Solution: Remove, align, and replace the paper.
9. **The paper is not being fed into the printer.**
   Cause: The paper is wrinkled.
   Solution: Remove the wrinkled paper from print tray.
   Cause: Printer set to print to a different paper size than currently loaded.
   Solution: Change paper size in print settings.
10. **Printer is printing incorrect colors.**
    Cause: Print cartridge is empty.
    Solution: Replace the printer cartridge.
    Cause: Print cartridge is defective.
    Solution: Replace the printer cartridge.
    Cause: Incorrect cartridge installed.
    Solution: Replace the printer cartridge.
    Cause: Print heads need to be cleaned and calibrated.
    Solution: Clean and calibrate the printer using the supplied software.

5) Mouse:

1) Loose Connection :

Cause and Solution:

- This is the most common problem in case of any mouse, be it PS/2 mouse, a USB mouse or a wireless mouse (its receiver).
- If you notice that the cursor is not responding to the mouse movements, check whether the connection between the mouse and the CPU has become loose.
- In case of a wireless mouse, check the connection of its receiver device. If it is loose, plug it in securely.

2) Accumulation of Dirt and Other Disturbance :

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Cause and Solution:

- If this is the problem you will have to open the chamber, and clean it as well as the trackball.
- Ideally you should use a clean cloth and alcohol to cleanse
- Also, an optical mouse may not work quite well if the surface below is shiny or transparent.

3) **Low Batteries :**

Cause and Solution:

- In case of wireless mouse, the problem may be to the batteries, interferences or range.
- Instead of concluding that there is a major problem and opting for a mouse repair, the foremost thing you need to do is to check if the batteries are low, and charge or replace them as required .
- Be sure to place the pointing device, i.e. mouse within the specified range of its receiver.

4) **Virus and Configuration Problem :**

Cause and Solution:

- Virus can also be one of the reasons for the mouse working abnormally i.e. if the cursor jumps across the screen uncontrollably, clicking on its own, etc.
- There are few viruses that may corrupt the mouse to malfunction.
- Mouse troubleshooting in this context will involves scanning the PC with a good antivirus in order to get rid of the virus.

5) **Problems with the Port or Mouse Driver :**

Cause and Solution:

- To verify whether the trouble is with the mouse or the PC, connect the mouse to a different PC to check.
- If it functions perfectly on another PC, then there is probably some problem with the port where it was connected or with the mouse driver in the system.
- Check by connecting the mouse to another port.

6) **Bluetooth Mouse Not Responding :**

Cause and Solution:

- Make sure that the polarity of the batteries is correct. The positive (+) and negative (-) ends of each battery must match the positive(+) and negative(-) connections in the battery housing.
- Make sure the batteries are charged.
- Verify that computer is Bluetooth enabled.
- Verify that the device drivers are installed.

6) Monitor

1) **Monitor Won't Turn On**

Solution

- Check if the monitor's power cable is plugged into a working outlet or power strip.
- Try using another power cable or electrical socket.
- Check and make sure that the power button isn't stuck.

2) **Dead Pixel**

Solution

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- Try applying pressure with your finger when you turn the monitor on and off. With any luck, this can help kick-start the electrical signal being sent to the pixel.
- If that doesn't work, try a program which rapidly changes the colours of the monitor.

3) The Monitor Is Already Turned On But There Is No Picture

Solution

- Check if the video cable is connected to the CPU firmly.
- If the monitor power indicator light is green but you still can't see a picture on the monitor, turn the computer off by pressing and holding the power button until it turns off. Once it's turned off, turn it right back on by pressing the power button on the CPU.
- If you suddenly a picture appears, you may be having an issue with the display settings in Windows. The resolution may be set too high or the monitor refresh rate may be set improperly.
- Start the system in "Safe Mode" and make any needed display setting changes that you may require. Start with the lowest resolution and refresh rate when you make the setting changes.
- Hold a lamp or flashlight right in front of the monitor. If you can see a picture in the background with the light on it, then the monitor's backlighting may be malfunctioning and the screen will need to be replaced.
- Connect the monitor to a different computer or CPU.
- If it still doesn't display a picture, replace the video cable.
- If it shows a picture, troubleshoot the video card of the first computer.

4) Image Color And Resolution Issues

Solution

- Try updating the driver.
- If that doesn't solve the problem, you may need a new video card entirely. This is quite common among old computers. Therefore, check if your computer is about to retire, because you just might consider buying a new PC altogether.

5) Video Light Is Too Dark Or Bright

Solution

- Reset the display settings to default.
- If you've made changes in the brightness and contrast controls and the screen still seems dark, you may need to take the monitor in for repair.

6) Images On The Screen Do Not Match The Screen Size

Solution

- Check your monitor's documentation to see the maximum resolution you can use.
- In Windows 10, right-click an open area in Desktop, select Display settings, select Advanced display settings, select the Resolution.

7) Flickering Monitor

Solution

- Try unplugging and firmly reattaching the cable to the video input and outputs.
- To reduce flicker on your monitor, try to increase the Refresh Rate in Windows.

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- In Windows 10, right-click the open area in Desktop, select Display settings, select Advanced display settings, select Display Adapter properties. Click Monitor tab, select the refresh rate that suits your monitor.

8) Monitor And Video Card Problem

Solution

- Reset the video card.
- Change the video.
- Replace the video card.
- Replace motherboard.

9) Crack Or Visible Spot On The Screen

Solution

- Unfortunately, if you see a huge crack or black spot on the screen, this indicates physical damage caused by blunt force. If you're distracted by the crack or if you cannot see much of the image anymore, your best solution is to replace it with a new monitor.
- It pays to know what to do when problems arise while using your monitor. If you think you have done every possible way to fix issues with your computer monitor, it may be time to send it to a technician to have it professionally checked and fixed.

❖ What is Preventive Maintenance and its Requirements.

- Preventive maintenance for a personal computer involves a series of routine tasks and checks to ensure its optimal performance, longevity, and security**.**

<u>Requirements :</u>

1) Software Updates:

- Regularly update the operating system(OS) and software applications to patch security vulnerabilities and ensure compatibility with the latest features and improvements.

2) Antivirus and Anti-malware Scans:

- Install and update antivirus software to protect against malicious threats. Schedule regular scans to detect and remove any potential malware.

3) Data Backup:

- Implement a reliable backup strategy to safeguard important data. Regularly back up files, documents, and other critical information to an external storage device or a cloud service.

4) Disk Cleanup:

- Periodically perform disk cleanup to remove temporary files, unnecessary system files, and cached data. This helps free up storage space and can improve system performance.

5) Disk Defragmentation:

- For traditional hard drives, run disk defragmentation to optimize file placement and improve disk access speed. This is typically done less frequently than other maintenance tasks.

6) Hardware Inspection:

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- Physically inspect the hardware components, including fans, heat sink, and power supplies. Ensure that there is no dust build up, and clean components as needed to prevent overheating.

7) **Update Device Drivers:**
   - Keep device drivers up to date, including graphics cards, network adapters, and other peripherals. Updated drivers can enhance performance, fix bugs, and improve compatibility.
8) **Check for Software Bloat:**
   - Regularly review installed programs and remove unnecessary or unused applications. This helps optimize system resources and reduces the risk of conflicts.
9) **Password Management:**
   - Regularly update and change passwords for user accounts. Use strong, unique password to enhance security.
10) **Monitor System:**
   - Utilize system monitoring tools to keep track of resource usage, temperatures, and overall system performance. This can help identify potential issues before they become serious problems.
11) **Network Security:**
   - Secure your network by updating router firmware, using strong encryption, and changing default login credentials. Regularly review and update network security settings.
12) **Review and Adjust Power Setting:**
   - Adjust power setting o balance energy efficiency with performance. This can help extend the lifespan of hardware components and reduce energy consumption.

❖ List out the step of Preventive Maintenance and described Preventive Maintenance.

- Cleaning your computer hardware.
- Downloading the latest updates for your computer software.
- Verifying you have the latest anti-virus protection updates on your computer.
- Running disk software utilities such as defrag and scandisk on your hard drive.
- Deleting unused programs or other files on your computer.
- If you don't turn off your computer it may be good to reboot your computer every few months, unless the computer is a critical computer such as a computer server.

➢ Preventive maintenance can be described as maintenance of equipment or systems before fault occurs. it is a daily maintenance design to retain the healthy condition of equipment and prevent failure through the prevention of deterioration, periodic inspection or equipment condition diagnosis, to measure deterioration. It divided in two sub groups

1) **Periodic maintenance (Time based maintenance-TBM)**
   - TBM consists of periodically inspecting, servicing and cleaning equipment and replacing parts to prevent sudden failure and process problems. periodic maintenance is planned based maintenance.
2) **Predictive maintenance**

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- This is a method in which the service life of important part is predicted based on inspection or diagnosis, in order to use the parts to the limit of their service life. Predictive maintenance is condition based maintenance. it manages trend values, by measuring and analysing data about deterioration and employs a surveillance system, designed to monitor conditions an on-line system.

❖ List the tools required for troubleshooting and maintenance.

1) **Multi-meter-** It is a device used to measure voltage, current and resistance in electronics & electrical equipment. it is also used to test continuity between 2 points to verify if there is any breaks in circuit or line .
2) **Digital multi-meter-** A digital multi-meter measures AC/DC voltage, resistance and current in an electric circuit. It is highly accurate and displays an LCD number readout.
3) **Oscilloscope –** An oscilloscope , previously called an oscillograph known as a CRO, or DSO, is a type of electronic test instrument that allows observation of constantly varying signal voltages.
4) **Logic analyzer-** It is an electronic instrument that captures and displays multiple signals from a digital system or digital circuit. Logic analyzer use for precise hardware troubleshooting, especially for timing issues.
5) **Logic probe-** It is a hand-held pen-like test probe used for analyzing and troubleshooting the logical states of a digital circuit. There are usually three differently-colours LEDs on the probe's body: Red and Green LEDs indicate high and low states respectively and amber LED indicates a pulse.
6) **Logic pulser-** The logic pulser is hand held tool used to inject pulses at the input of a gate under test. It maps single or stream of pulses at different frequencies.
7) **Current tracer –** The current tracer is hand held tool which detects current flow in electronics circuits. It is used to locate shorted components, track shorts, solder bridge, ground short etc.

❖ Explain Diagnostic software in details.

- Diagnostics software is a powerful tool for testing and troubleshooting your computer hardware. It can help you identify faulty components, monitor performance, optimize settings, and fix errors.

1) MemTest:
   - Mem test is a simple test that determines whether your computer can reliably can store information in its RAM. with this tool, you can define how much memory to test and then run the test. If an error is reported, your RAM is suspect. The only drawback to memtest is that it can't pinpoint which memory chip is bad. But if you're looking for a memory tester that can be run while windows itself is running, Memtest is your tool.
2) CCleaner:
   - CCleaner can help you clean out of your machine. Not only can CCleaner keep your disk cache clean. It can keep your registry free of errors. The result in a computer running efficiently. Using CCleaner regularly will go a long way toward keeping your machines running well.
3) HDD Scan:

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- HDD scan is a free hard disk diagnostic tool that supports standard drives, RAID arrays, Flash USB, and SSD drives. With this tool, you can scan for errors. By regularly using this tool, you may avoid a avoid disaster.

4) Defragged:
   - Defragged is one of the best defrag programs available. It will defrag an entire hard disk as well as individual files or folders. You can specify what you want to defrag and how you want to defrag it.
   - Defragged also offers a portable version, so you don't have to install the tool on every machine you touch.
   - Defragged setting up regularly scheduled defrags to keep every machine on your network running smoothly.

❖ List out Laymen Check.

- **Look for damage**- check for any physical damage to the computer, cables, keyboard, mouse, etc.
- **Clean dust**- Use compressed air to below out dust from inside the PC case, fans, vents. Dust causes overheating.
- **Check connections**- make sure all cables and cords are plugged in tightly. Replace any frayed or damaged cables.
- **Update software**- keep your operating system, antivirus, programs , and device drivers up to date. Updates fix bugs and improve security.
- **Check storage space**- don't let your hard drive fill up. Delete unused files and run disk cleanup to free up space.
- **Back up files**- copy important files onto an external drive or cloud storage to avoid losing data.
- **Scan for viruses**- run antivirus software regularly to check for and remove malware.
- **Check performance**- use task manager or activity monitor to check for software conflicts or start up programs slowing your PC.
- **Test internet speed**- run speed tests to catch internet connection issues. Reboot router if speed is consistently slow.
- **Update drivers**- keep hardware drivers like graphics, printer, network cards updated for better stability.
- **Check peripherals**- make sure keyboards, printers, scanners and other peripherals are working right.
- **Monitor temperature**- use software to check CPU and GPU temperatures stay in safe range.
- **Check battery-** for laptop, monitor battery capacity and lifespan. Replace battery every few years.
- **Fix errors**- check system logs for hardware/ software errors. Fix any issues.
- **Diagnose problems**- run built-in diagnostic tools to check for issues.

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❖ Explain steps of dissembling PC Components.

1) **Step-1 Unplugging**
   - Unplugging the power Cord From the PC
   - Unplug all the Peripherals attached to the computer, such as the keyboard, mouse, monitor, headphones.
2) **Step-2** Open the case first remove the screws of the left side cover and slide the side cover.
3) **Step-3 Disconnect all the connectors**
   - Disconnect all the connected to the motherboard.
   - These include SATA power cable and data cable of HDD as well as SATA cable of optical drive.
4) **Step-4 Remove the fan**
   - Remove the fan now
5) **Step-5 Remove the power supply**
   - Remove the power supply is connected to the motherboard by A 20-pin connector and 4-pin Connector.
6) **Step-6 Remove HDD and optical drive**
   - Remove the SATA cable connecting to the HDD and Optical Drive in Motherboard.
7) **Step-7 Remove RAM Modules**
   - To remove the RAM
8) **Step-8 Remove Expansion cards**
   - To remove the expansion cards, disconnected the cables attached to it.
9) **Step-9 Remove Motherboard**
   - To remove the motherboard , disconnected all the cables from the motherboard.

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